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Introduction

1. About the work book:

This workbook is intended to be used by T.Y.B.Sc. (Computer Science) students for the USCSP 369 Object Oriented Programming using Java II Laboratory Assignments in Semester–VI. This workbook is designed by considering all the practical concepts / topics mentioned in syllabus.

2. The objectives of this workbook are:

- 1) Defining the scope of the course.
- 2) To have continuous assessment of the course and students.
- 3) Providing ready reference for the students during practical implementation.
- 4) Provide more options to students so that they can have good practice before facing the examination.
- 5) Catering to the demand of slow and fast learners and accordingly providing the practice assignments to them.

3. How to use this workbook:

- 1) The workbook is divided into five assignments.
- 2) Each Java assignment has three SETs.
- 3) It is mandatory for students to complete the SET A and SET B in given slot. Set C prepares the students for the viva in the subject. Students should spend additional time either at home or in the Lab and solve these problems so that they get a deeper understanding of the subject.

4. Instructions to the students

- 1) Students are expected to carry this book every time they come to the lab for practical.
- 2) Students should prepare oneself beforehand for the Assignment by reading the relevant material.
- 3) Instructor will specify which problems to solve in the lab during the allotted slot and student should complete them and get verified by the instructor. However student should spend additional hours in Lab and at home to cover as many problems as possible given in this work book.
- 4) Students will be assessed for each exercise on a scale from 0 to 5.

Not done	0
Incomplete	1
Late Complete	2
Needs improvement	3
Complete	4
Well Done	5

5. Guidelines for Instructors

- 1) Explain the assignment and related concepts in around ten minutes using white board if required or by demonstrating on Projector.
- 2) You should evaluate each assignment carried out by a student on a scale of 5 as specified above by ticking appropriate box.
- 3) The value should also been written on assignment completion page of the respective Lab course.

6. Guidelines for Lab administrator

You have to ensure appropriate hardware and software is made available to each student.

The operating system and software requirements on server side and also client-side areas given below:

- Operating system: Linux
- Editor: Any linux based editor like vi, gedit, eclipse etc.
- Compiler: javac (Note : JAVA 8 and above version)
- Web server: Tomcat
- Database: PostgreSQL

Assignment Completion Sheet

Sr. No	Assignment Name	Marks (Out of 5)	Signature
1	Collections		
2	Multithreading		
3	Database Programming		
4	Servlets		
5	JSP		
Total out of 25			
Total (Out of 10)			

CERTIFICATE

This is to certify that Mr./Ms. _____
has successfully completed **TYBSc (Computer Science) USCSP**
369 Practical course based on USCS 365 Object Oriented
Programming using Java II Semester VI course in year
_____ and his/her seat no. is _____.
He / She have scored _____ marks out of 10.

Instructor

HOD/Coordinator

Internal Examiner

External Examiner

Assignment 1: Collections

Objectives

- Study the Collections framework in java
- To store and manipulate group of objects
- Use various collections

What is Collection?

Collection is a group of objects.

Collection is a container object. It is used for storing multiple homogeneous and heterogeneous, unique and duplicate objects without size limitation.

What is the need of Collection?

In Java we can store and transfer the data in the following ways.

1. Primitive data types

We can use primitive data types for storing only one element data and only one type of data.

2. Class objects

Can store multiple fixed number of values of different type and can store multiple data elements of multiple types.

3. Array Object

Can store multiple fixed number of values of same type for storing many values of same data type.

4. Collection Object

Can store multiple objects of same and different types without size limitations
Thus, if our requirement is store and process multiple objects then we go for Collection Framework.

Limitations of Arrays -

1. Type (It is homogeneous in nature)
2. Size
3. Storing order
4. Operation Problem

Arrays Vs Collections

Arrays	Collection
Fixed in size	Growable
Can hold only homogeneous data	Can hold both homogeneous & heterogeneous data
No predefined method support	For every requirement methods are available
Arrays can hold both primitives and objects	Collections can hold only objects

Collection Framework

- The Collection Framework in Java is a collection of interfaces and classes to store, process and transfer the data efficiently.
- Collections are growable in nature. i.e., based on run time requirement we can store any number of elements.
- Collections can hold both homogeneous and heterogeneous data elements.
- We can transfer the data from one method to another of any type and any number of elements.
- For every requirement ready made method support is available. Hence being a programmer, we just have to know how to use the predefined methods
- A collection framework provides built-in interfaces, classes and methods which we can use to directly create and use a collection instead of writing long code manually.
- A collections framework is a unified architecture for representing and manipulating collections.
- All collections frameworks contain the following

Collection

It is at the top of collection hierarchy and must be implemented by any class that defines a collection. Following are some of the commonly used methods in this interface.

Methods	Description
boolean add(Object e)	Used to add objects to a collection. Returns true if obj was added to the collection. Returns false if obj is already a member of the collection, or if the collection does not allow duplicates.
boolean addAll(Collection C)	Add all elements of collection C to the invoking collection. Returns true if the element were added. Otherwise, returns false.
boolean remove(Object obj)	To remove an object from collection. Returns true if the element was removed. Otherwise, returns false.
boolean removeAll(Collection C)	Removes all element of collection C from the invoking collection. Returns true if the collection's elements were removed. Otherwise, returns false.
boolean contains(Object obj)	To determine whether an object is present in collection or not. Returns true if obj is an element of the invoking collection. Otherwise, returns false.
boolean isEmpty()	Returns true if collection is empty, else returns false.
int size()	Returns number of elements present in collection.
void clear()	Removes total number of elements from the collection.

Object[] toArray()	Returns an array which consists of the invoking collection elements.
boolean retainAll(Collection c)	Deletes all the elements of invoking collection except the specified collection.
Iterator iterator()	Returns an iterator for the invoking collection.
boolean equals(Object obj)	Returns true if the invoking collection and obj are equal. Otherwise, returns false.
Object[] toArray(Object array[])	Returns an array containing only those collection elements whose type matches of the specified array.

Implementations

The general-purpose implementations are summarized in the following table.

General – Purpose Implementation					
Interface			Implementation		
	Hash table	Resizable Array	Tree	Linked List	Hash table + Linked List
Set	HashSet		TreeSet		LinkedHashSet
List		ArrayList		LinkedList	
Map	HashMap		TreeMap		LinkedHashMap

ArrayList class

The ArrayList class implements the List interface. It uses a dynamic array to store the duplicate element of different data types. The ArrayList class maintains the insertion order and is non-synchronized. The elements stored in the ArrayList class can be randomly accessed. .

Important points to note

- Underlying data structure for ArrayList is Resizable Array.
- Duplicates are allowed.
- Insertion order is preserved.
- Heterogeneous objects insertion is allowed.
- 'Null' insertion is possible.
- ArrayList is best choice if our frequent operation is retrieval.
- ArrayList is worst choice if our frequent operation is insertion or deletion in middle

Methods of ArrayList

Method	Description
void add(int index, Object element)	It is used to insert the specified element at the specified position in a list.
boolean addAll(int index, Collection c)	It is used to append all the elements in the specified collection, starting at the specified position of the list.
void clear()	It is used to remove all of the elements from this list.

void ensureCapacity(int requiredCapacity)	It is used to enhance the capacity of an ArrayList instance.
get(int index)	It is used to fetch the element from the particular position of the list.
boolean isEmpty()	It returns true if the list is empty, otherwise false.
boolean contains(Object o)	It returns true if the list contains the specified element
boolean remove(Object o)	It is used to remove the first occurrence of the specified element.
boolean removeAll(Collection c)	It is used to remove all the elements from the list.
void replaceAll(Collection c)	It is used to replace all the elements from the list with the specified element.
void retainAll(Collection c)	It is used to retain all the elements in the list that are present in the specified collection.
int size()	It is used to return the number of elements present in the list.

Linked List

Java LinkedList class provides implementation of linked-list data structure.

Important points to note

- It uses doubly linked list to store the elements.
- Duplicates are allowed.
- Insertion order is preserved.
- Heterogeneous objects insertion allowed.
- 'Null' insertion is possible.
- .LinkedList is best choice if our frequent operation is insertion or deletion in middle.
- LinkedList is worst choice if our frequent operation is retrieval operation.

Methods of Java LinkedList

Method	Description
void add(int index, Object element)	It is used to insert the specified element at the specified position index in a list.
boolean addAll(Collection c)	It is used to append all of the elements in the specified collection to the end of this list, in the order that they are returned by the specified collection's iterator.
void addFirst(Object element)	It is used to insert the given element at the beginning of a list.
void addLast(Object element)	It is used to append the given element to the end of a list.
void clear()	It is used to remove all the elements from a list.
boolean contains(Object o)	It is used to return true if a list contains a specified element.
Object element()	It is used to retrieve the first element of a list.
Object get(int index)	It is used to return the element at the specified position in a list.
Object peek()	It retrieves the first element of a list
void push(Object e)	It pushes an element onto the stack represented by a list.

Object remove()	It is used to retrieve and removes the first element of a list.
Object remove(int index)	It is used to remove the element at the specified position in a list.
int size()	It is used to return the number of elements in a list.

Set Implementation

Sets are further classified into the following:

- **HashSet**
- **LinkedHashSet**
- **TreeSet**

HashSet

- Underlying data structure for HashSet is hash table.
- Duplicates are not allowed
- Insertion order is not preserved
- Heterogeneous objects insertion is allowed
- 'Null' insertion is possible
- HashSet is best choice if our frequent operation is search

LinkedHashSet

- Underlying data structure for LinkedHashSet is hash table and Linked List.
- Duplicates are not allowed.
- Insertion order is not preserved.
- Heterogeneous objects insertion is allowed.
- 'Null' insertion is possible.
- LinkedHashSet is best choice to develop cache based application.

TreeSet

- Underlying data structure is balanced tree.
- Duplicates are not allowed
- Insertion order is not preserved but all objects will be inserted according to some sorting order
- Heterogeneous objects insertion is not allowed
- 'Null' insertion is possible but only once
- HashSet is best choice if our frequent operation is search

Map

A Map is useful if you have to search, update or delete elements on the basis of a key.

There are two interfaces for implementing Map in java: **Map** and **SortedMap**, and

Three classes: **HashMap**, **LinkedHashMap**, and **TreeMap**.

Class	Description
HashMap	HashMap is the implementation of Map, but it doesn't maintain any order.

LinkedHashMap	LinkedHashMap is the implementation of Map. It inherits HashMap class. It maintains insertion order.
TreeMap	TreeMap is the implementation of Map and SortedMap. It maintains ascending order.

HashMap

Map contains its own methods. collection terminology is not applicable

- Underlying datastructure forHashMapis hash table
- Duplicates keys are not allowed but values are allowed
- Insertion order is not preserved and it is based on hashcode of keys
- Heterogeneous keys and values are allowed
- ‘Null’ insertion is possible
- Best choice for searching

LinkedHashMap

- Underlying datastructure forLinkedHashMap is hash table and Linked List
- Duplicates keys are not allowed but values are allowed
- Insertion order is preserved and it is based on hashcode of keys
- Heterogeneous keys and values allowed
- ‘Null’ insertion is possible
- Best choice for searching

SortedMap

It is child interface of Map.

- Underlying data structure for SortedMap is hash table
- Duplicates keys are not allowed but values
- Insertion order is not preserved and it is based on hashcode of keys
- Heterogeneous keys and values allowed
- ‘Null’ insertion is possible
- Best choice for searching

TreeMap

- Underlying data structure is RED-BLACK TREE
- Duplicates keys are not allowed but values can be duplicated
- Insertion order is not preserved and it is based on some sorting order ofkeys
- Heterogeneous keys and values not allowed
- Null acceptance is not there

Cursors

A Java Cursor is an Iterator, which is used to iterate or traverse or retrieve a Collection or Stream object’s elements one by one.

Java supports the following three different cursors.

- **Enumeration**
- **Iterator**
- **ListIterator**

Enumeration

We can use Enumeration to get objects one by one from the legacy collection objects.

We can create Enumeration object by using `elements()` method.

Enumeration interface defines the following two methods

```
public boolean hasMoreElements();
```

```
public Object nextElement();
```

Iterator

We can apply Iterator concept for any collection object hence it is universal cursor. By using this we can perform both read and remove operations. We can create Iterator by using `Iterator()` of collection interface.

```
public Iterator iterator();
```

```
Iterator itr = c.Iterator();
```

Where c is any collection object

Iterator Methods

Methods	Description
<code>next()</code>	Returns the next object
<code>boolean hasNext()</code>	This returns a true value if a high number of elements are encountered during iteration.
<code>remove()</code>	This method removes the current element. Throws <i>IllegalStateException</i> if an attempt is made to call <code>remove()</code> that is not preceded by a call to <code>next()</code> .

ListIterator

By using this we can move either to f/w or b/w direction and hence it is bidirectional cursor. We can perform replacement and addition of new objects in addition to read and remove operation.

Note - It is most powerful cursor but its limitation is, it is applicable only for list implemented class objects and it is not universal cursor.

ListIterator methods

Methods	Description
<code>void add(Object obj)</code>	Inserts obj into the list in front of the element that will be returned by the next call to <code>next()</code> .
<code>boolean hasNext()</code>	Returns true if there is the next element. Otherwise, returns false.
<code>boolean hasPrevious()</code>	Returns true if there is a previous element. Otherwise, returns false.
<code>Object next()</code>	Returns the next element. A <i>NoSuchElementException</i> is thrown if there is not the next element.
<code>int nextIndex()</code>	Returns the index of the next element. If there is not the next element, returns the size of the list.
<code>Object previous()</code>	Returns the previous element. A <i>NoSuchElementException</i> is thrown if there is not a previous element.
<code>int previousIndex()</code>	Returns the index of the previous element. If there is not a previous element, returns -1.
<code>void remove()</code>	Removes the current element from the list. An <i>IllegalStateException</i> is thrown if <code>remove()</code> is called before <code>next()</code> or <code>previous()</code> is invoked.

void set(Object obj)	Assigns obj to the current element. This is the element last returned by a call to either next() or previous().
----------------------	---

Comparator

- Comparator interface is used to order the objects of a user-defined class.
- This interface is found in java.util package and contains 2 methods **compare(Object obj1, Object obj2)** and **equals(Object element)**.
- It provides multiple sorting sequences, i.e., you can sort the elements on the basis of any data member, for example, rollno, name, age or anything else

Methods of Java Comparator Interface

There are two methods of Comparators in, namely:

Methods	Description
compare(Object obj1, Object obj 2)	Compares the first object with another
equals(Object obj)	Compares current object with specified obj

Self-Activity

Note: To use any Collection class in your program, you need to import java.util package.

Whenever you print any Collection class, it gets printed inside the square brackets [] with its elements.

Sample Program 1 : Program to demonstrate ArrayList

```

/* Program to demonstrate ArrayList*/

import java.util.ArrayList;
class ArrayListDemo
{
    public static void main(String[] args)
    {
        // creating an Array List named colors
        ArrayList Al = new ArrayList();
        // add elements in the
        Array ListAl.add("Red");
        Al.add(5);
        Al.add("Null");
        Al.add("Orange");
        Al.add("Red");
        // printing the ArrayList
        System.out.println(Al);
    }
}

```

Sample Program 2 : Program to demonstrate LinkedList

```
/* Program to demonstrate LinkedList */

import java.util.*;
public class LinkedList1{
    public static void main(String args[]){
        LinkedList<String> al=new LinkedList<String>();
        al.add("Ravi");
        al.add("Vijay");
        al.add("Ravi");
        al.add("Ajay");
        Iterator<String>itr=al.iterator();
        while(itr.hasNext()){
            System.out.println(itr.next());
        }
    }
}
```

Sample Program 3 : Program to demonstrate HashSet

```
/* Program to demonstrate HashSet */
import java.util.HashSet;

public class Main {
    public static void main(String[] args) {
        // Create HashSet object

        HashSet hs = new HashSet(5, 0.5f);
        System.out.println(hs.add("one"));
        System.out.println(hs.add("two"));
        System.out.println(hs.add("three"));
        System.out.println(hs.add("four"));
        System.out.println(hs.add("five"));
        // Print out the HashSet object
        System.out.println(hs);
        // Add a duplicate item to the HashSet
        Boolean b = hs.add("one");
        System.out.println("Duplicate item allowed = " + b);
        System.out.println(hs);

    }
}
```

Sample Program 4 : Program to demonstrate LinkedHashSet

```
/* Program to demonstrate LinkedHashSet */
import java.util.*;
public class Test4{
    public static void main(String args[]){
        LinkedHashSet<String> set=new
        LinkedHashSet<String>();set.add("Java");
        set.add("ML");
        set.add("Python");
        set.add("AI");
        Iterator<String>itr=set.iterator();
        while(itr.hasNext()){
            System.out.println(itr.next());
        }
    }
}
```

Sample Program5 :Program to demonstrate TreeSet

```
/* Program to demonstrate TreeSet*/
import java.util.Set;
import java.util.TreeSet;
public class Main {
    public static void main(String[] args) {
        Set ts = new TreeSet();
        ts.add("one");
        ts.add("two");
        ts.add("three");
        ts.add("four");
        ts.add("three");
        System.out.println("Members from TreeSet = " + ts);
    }
}
```

Sample Program8 : Program to demonstrate ListIterator

```
/* Program to demonstrate ListIterator */
import java.util.ArrayList;
import java.util.ListIterator;
public class Main {
    public static void main(String[] args) {
        // Create ArrayList object with capacity of 2 elements
        ArrayList al = new ArrayList(2);
        System.out.println(al+"", size = "+al.size());
        // Add items to the ArrayList
        al.add("R");
        al.add("U");
        al.add("O");
        al.add(new String("x"));
        al.add(2, new Integer(10));
        System.out.println(al+"", size = " + al.size());
        // Remove item
        al.remove("U");
        System.out.println(al+"", size = " + al.size());
        // Check if the list contains the specified element
        Boolean b = al.contains("x");
        System.out.println("The list contains x = " + b);
        b = al.contains("p");
        System.out.println("The list contains p = " + b);
        b = al.contains(new Integer(10));
        System.out.println("The list contains Integer of 10 = " + b);
        // Create ListIterator and iterate entries in it
        ListIterator li = al.listIterator();
        while (li.hasNext())
            System.out.println("From ListIterator = " + li.next());

        // Create Object array from ArrayList
        Object a[] = al.toArray();
        for (int i=0; i<a.length; i++)
            System.out.println("From an Array = " + a[i]);
    }
}
```

Lab Assignments

SET A

1. Write a java program to accept names of 'n' cities, insert same into array list collection and display the contents of same array list, also remove all these elements.
2. Write a java program to read 'n' names of your friends, store it into linked list, also display contents using ListIterator interface.
3. Accept n integers from the user and store them in a collection. Display them in the sorted order. The collection should not accept duplicate elements. (Use a suitable collection). Search for a particular element using predefined search method in the Collection

framework.

4. Create a Hash map containing Employee name and Salary. Display the details of the hash map. Also search for a specific Employee and display Salary of that Employee.

SET B

1. Construct a linked List containing names of colours: red, blue, yellow and orange. Then extend your program to do the following:
 - a. Display the contents of the List using an Iterator
 - b. Display the contents of the List in reverse order using a ListIterator
 - c. Create another list containing pink and green. Insert the elements of this list between blue and yellow.
2. Create a java application to store city names and their STD codes using an appropriate collection. The GUI should allow the following operations:
 - a. Add a new city and its code (No duplicates)
 - b. Remove a city from the collection
 - c. Search for a city name and display the code
3. Write a menu driven program to create link list of integer objects. Do the following:
 - i. Add element at first position
 - ii. Delete last element
 - iii. Display the size of link list

SET C

1. Read a text file, specified by the first command line argument, into a list. The program should then display a menu which performs the following operations on the list:
 1. Insert line 2. Delete line 3. Append line 4. Modify line 5. ExitWhen the user selects Exit, save the contents of the list to the file and end the program.
2. Write a program that loads names and phone numbers from a text file where the data is organized as one line per record and each field in a record are separated by a tab (\t).it takes a name or phone number as input and prints the corresponding other value from the hash table (hint: use hash tables)

Assignment Evaluation

0: Not Done		1: Incomplete		2: Late Complete	
3: Needs Improvement		4: Complete		5: Well Done	

Signature of the instructor: _____ **Date:** _____

Assignment 2: Multithreading

Objectives

- To create and use threads in java
- To demonstrate multithreading

Introduction:

A program can be divided into a number of small processes. Each small process can be addressed as a single thread (a lightweight process).

Multithreaded programs contain two or more threads that can run concurrently and each thread defines a separate path of execution. This means that a single program can perform two or more tasks simultaneously. For example, one thread is writing content on a file at the same time another thread is performing spelling check.

Why use Threads in Java

The Java run-time system depends on threads for many things. Threads reduce inefficiency by preventing the waste of CPU cycles.

Why Multithreading?

Thread has many advantages over the process to perform multitasking. Process is heavy weight, takes more memory and occupy CPU for longer time that may lead to performance issue with the system. To overcome this issue process is broken into small unit of independent sub-process. These sub-process are called threads that can perform independent task efficiently. So nowadays computer systems prefer to use thread over the process and use multithreading to perform multitasking.

The main thread

When we run any java program, the program begins to execute its code starting from the main method. Therefore, the JVM creates a thread to start executing the code present in main method. This thread is called as main thread. Although the main thread is automatically created, you can control it by obtaining a reference to it by calling `currentThread()` method. Two important things to know about main thread are,

- It is the thread from which other threads will be produced.
- It must be always the last thread to finish execution

How to Create a Java Thread

Java lets you create thread in following two ways:-

1. By implementing the Runnable interface
2. By extending the Thread

Method 1: Thread creation by extending Thread class

```
class MultithreadingDemo extends Thread
{
    public void run()
    {
        System.out.println("My thread is in running state.");
    }
    public static void main(String args[])
    {
        MultithreadingDemo obj=new MultithreadingDemo();
        obj.start();
    }
}
```

Output:

My thread is in running state.

Method 2: Thread creation by implementing Runnable Interface

```
class MultithreadingDemo implements Runnable
{
    public void run()
    {
        System.out.println("My thread is in running state.");
    }
    public static void main(String args[])
    {
        MultithreadingDemo obj=new MultithreadingDemo();
        Thread tobj =new Thread(obj);
        tobj.start();
    }
}
```

Output:

My thread is in running state.

Important methods of Thread class

Method	Description
setName()	To give thread a name
getName()	Return thread's name
getPriority()	Return thread's priority
isAlive()	Checks if thread is still running or not
join()	Wait for a thread to end
run()	Entry point for a thread
sleep()	Suspend thread for a specified time
start()	Start a thread by calling run() method
activeCount()	Returns an estimate of the number of active threads in the current thread's thread group and its subgroups.

checkAccess()	Determines if the currently running thread has permission to modify this thread.
currentThread()	Returns a reference to the currently executing thread object.
getId()	Returns the identifier of this Thread.
getState()	Returns the state of this thread.
interrupt()	Interrupts this thread.
isAlive()	Tests if this thread is alive.
isDaemon()	Tests if this thread is a daemon thread.
isInterrupted()	Tests whether this thread has been interrupted.
setPriority(int newPriority)	Changes the priority of this thread.
yield()	A hint to the scheduler that the current thread is willing to yield its current use of a processor.

Priority of a Thread

Each thread have a priority. Priorities are represented by a number between 1 and 10.

Three constants defined in Thread class:

```
public static int MIN_PRIORITY
public static int NORM_PRIORITY
public static int MAX_PRIORITY
```

Default priority of a thread is 5 (**NORM_PRIORITY**). The value of **MIN_PRIORITY** is 1 and the value of **MAX_PRIORITY** is 10.

Thread Synchronization

- If multiple threads are trying to operate simultaneously on same java object, then there may be a chance of data inconsistency problem. To overcome this problem, we should go for Synchronization
- If a method or block declared as synchronized then at a time only one thread is allowed to execute that method or block on the given object so that data inconsistency prob will be resolved.

Inter Thread Communication in Java

- Inter-thread communication in Java is a technique through which multiple threads communicate with each other
- There are several situations where communication between threads is important. For example, suppose that there are two threads A and B. Thread B uses data produced by Thread A and performs its task.
- If Thread B waits for Thread A to produce data, it will waste many CPU cycles. But if threads A and B communicate with each other when they have completed their tasks, they do not have to wait and check each other's status every time.
- Inter thread communication in Java can be achieved by using three methods
 1. **wait()**
 2. **notify()**
 3. **notifyAll()**
- Note-These methods can be called only from within a synchronized method or synchronized block of code

Self Activity

Sample Program1: Below is a program that illustrates instantiation and running of threads using the Runnable interface.

```
class RunnableThread implements Runnable
{
    Thread runner;
    public RunnableThread()
    {
    }
    public RunnableThread(String threadName)
    {
        runner = new Thread(this, threadName); // Create a new thread.
        System.out.println(runner.getName());
        runner.start(); // Start the thread.
    }
    public void run()
    {
        //Display info about this particular thread
        System.out.println(Thread.currentThread());
    }
}
public class RunnableExample
{
    public static void main(String[] args)
    {
        Thread thread1 = new Thread(new RunnableThread(), "thread1");
        Thread thread2 = new Thread(new RunnableThread(), "thread2");
        RunnableThread thread3 = new RunnableThread("thread3");
        //Start the threads
        thread1.start();
        thread2.start();
        try
        {
            //delay for one second Thread.currentThread().sleep(1000);
        }
        catch (InterruptedException e)
        {
        }
        //Display info about the main thread
        System.out.println(Thread.currentThread());
    }
}
```

Sample Program2:Creating multiple threads using the Thread class.

```
class MyThread extends Thread
{
String message;
MyThread(String message)
{
this.message = message;
}
public void run()
{
try
{
for(int i=1; i<=5; i++)
{
System.out.println(message + "-" + i);
Thread.sleep(5000); //sleep for 5 seconds
}
}
catch(InterruptedException ie) { }
}
}
public class MultipleThreadDemo
{
public static void main( String[] args)
{
MyThread t1 = new MyThread("One");
MyThread t2 = new MyThread("Two");
System.out.println(t1);
System.out.println(t2); t1.start();
t2.start();
}
}
```

Sample Program3: Demonstrating Priority of a Thread

```
class PriorityDemo extends Thread
{
    public void run()
    {
        System.out.println("running thread name is:"+Thread.currentThread().getName());
        System.out.println("running thread priority is:"+Thread.currentThread().getPriority());
    }
    public static void main(String args[])
    {
        PriorityDemo m1=new PriorityDemo();
        PriorityDemo m2=new PriorityDemo();
        m1.setPriority(Thread.MIN_PRIORITY);
        m2.setPriority(Thread.MAX_PRIORITY);
        m1.start();
        m2.start();
    }
}
```

Sample Program4 : Thread Synchronization

```
class mythread extends Thread
{
    String msg[]={"Java", "Supports", "Multithreading", "Concept"};
    mythread(String name)
    {
        super(name);
    }
    public void run()
    {
        display(getName());
        System.out.println("Exit from "+getName());
    }
    synchronized void display(String name ) //Synchronized method
    {
        for(int i=0;i<msg.length;i++)
        {
            System.out.println(name+msg[i]);
        }
    }
}
```

```

    /* Main class */
class MySynchro
{
    public static void main(String args[])
    {
        mythread t1=new mythread("Thread 1: ");
        mythread t2=new mythread("Thread 2: ");
        t1.start(); t2.start();
        System.out.println("Main thread exited");
    }
}

```

Sample Program5 : Demonstrating Inter-thread communication of a Thread

```

class SampleThread extends Thread
{
    int tBal = 0;
    public void run()
    {
        synchronized (this)
        {
            System.out.println("Thread calculation for total balance");
            for (int i = 0; i <= 30; i++)
            {
                tBal = tBal + i;
            }
            System.out.println("Thread gives notification call");
            this.notify();
        }
    }
}

public class DemoThread
{
    public static void main(String[] args) throws InterruptedException
    {
        SampleThread st = new SampleThread ();
        st.start();
        synchronized (st)
        {
            System.out.println("Thread calling wait() Method");
            st.wait();
            System.out.println("Thread got notification");
            System.out.println("Total Balance " + st.tBal);
        }
    }
}

```

Lab Assignments

SET A

1. Write a java program to display name and priority of a Thread. Change the priority of thread to 7.
2. Program to define a thread for printing text on output screen for 'n' number of times. Create 3 threads and run them. Pass the text 'n' parameters to the thread constructor. Example:
 - a. First thread prints "COVID19" 10 times.
 - b. Second thread prints "LOCKDOWN2020" 20 times
 - c. Third thread prints "VACCINATED2021" 30 times
3. Write a program in which thread sleep for 2 sec in the loop in reverse order from 100 to 1 and change the name of thread.
4. Write a program that implements a multi-thread application that has three threads. First thread generates random integer after every 1 second and if the value is even, second thread computes the square of the number and prints. If the value is odd, the third thread will print the value of cube of the number.

SET B

1. Write a program to calculate the sum and average of an array of 1000 integers (generated randomly) using 10 threads. Each thread calculates the sum of 100 integers. Use these values to calculate average. [Use join method].
2. Write a Java program using Runnable interface to blink Text on the frame.
3. Write a Multithreading program in java to display the number's between 1 to 100 continuously in a TextField by clicking on button. (Use Runnable Interface).
4. Write a program for a simple search engine. Accept a string to be searched. Search for the string in all text files in the current folder. Use a separate thread for each file. The result should display the filename, line number where the string is found.
5. Write a program to solve producer consumer problem in which a producer produces a value and consumer consume the value before producer generate the next value. (Hint: use thread synchronization)

SET C

1. Write a Java program to show lifecycle (creation, sleep, and dead) of a thread. Program should print randomly the name of thread and value of sleep time. The name of the thread should be hard coded through constructor. The sleep time of a thread will be a random integer in the range 0 to 4999.

2. Write a program to create a thread for moving a ball inside a panel vertically. The ball should be created when the user clicks on the start button.

3. Write a program that simulates a traffic light. The program lets the user select one of three lights: red, yellow, or green with radio buttons. On selecting a button, an appropriate message with “stop” or “ready” or “go” should appear above the buttons in a selected color. Initially there is no message shown.

Assignment Evaluation

0: Not Done		1: Incomplete		2: Late Complete	
3: Needs Improvement		4: Complete		5: Well Done	

Signature of the instructor: _____ **Date:** _____

Assignment 3: Database Programming

Objectives

- To communicate with a database using java.
- To execute queries on tables.
- To obtain information about the database and tables

JDBC : Java Database Connectivity

This API contains of a set of **classes** and **interfaces** to enable programmers to communicate with a database using java. These classes and interfaces are in the java.sql package.

The JDBC API makes it possible to do three things:

- i. Establish a connection with a data source.
- ii. Send queries and update statements to the data source.
- iii. Process the results.

Classes Name	Description
DriverManager	The basic service for managing a set of JDBC drivers.
DriverPropertyInfo	Driver properties for making a connection
Date	Represents an SQL DATE value.
SQLPermission	The permission for which the SecurityManager will check when code that is running in an applet calls the DriverManager.setLogWriter method or the DriverManager.setLogStream (deprecated) method.
Time	Represents an SQL TIME value.
Timestamp	Represents an SQL TIMESTAMP value.
Types	Defines constants that are used to identify generic SQL types, called JDBC types.

JDBC Drivers

To communicate with a database, you need a database driver.

There are four types of drivers :

1. Type 1: JDBC-ODBC Bridge driver.
2. Type 2: Native-API partly-Java driver.
3. Type 3: JDBC-Net pure Java driver.
4. Type 4: Native-protocol pure Java driver.

Load Driver

For postgresql, use the driver :

org.postgresql.Driver

To load the driver, use the following command :

Syntax :

Class.forName("DiverName");

Example :

```
Class.forName("org.postgresql.Driver");
```

Establishing a connection

To establish a connection with the database, use the `getConnection` method of the `DriverManager` class.

This method returns a `Connection` object.

```
DriverManager.getConnection("url", "user", "password");
```

Example :

```
Connection conn = DriverManager.getConnection ("jdbc:postgresql:TestDB", "postgres", "");
```

Methods of Connection class:

Methods	Description
<code>void close()</code>	Releases this <code>Connection</code> object's database and JDBC resources immediately instead of waiting for them to be automatically released.
<code>void commit()</code>	Makes all changes made since the previous commit/rollback permanent and releases any database locks currently held by this <code>Connection</code> object.
<code>Statement createStatement()</code>	Creates a <code>Statement</code> object for sending SQL statements to the database.
<code>Statement createStatement(int resultSetType, int resultSetConcurrency)</code>	Creates a <code>Statement</code> object that will generate <code>ResultSet</code> objects with the given type and concurrency.
<code>Boolean getAutoCommit()</code>	Retrieves the current auto-commit mode for this <code>Connection</code> object.
<code>DatabaseMetaData getMetaData()</code>	Retrieves a <code>DatabaseMetaData</code> object that contains metadata about the database to which this <code>Connection</code> object represents a connection.
<code>CallableStatement prepareCall(String sql)</code>	Creates a <code>CallableStatement</code> object for calling database stored procedures.
<code>CallableStatement prepareCall(String sql, int resultSetType, int resultSetConcurrency)</code>	Creates a <code>CallableStatement</code> object that will generate <code>ResultSet</code> objects with the given type and concurrency.
<code>PreparedStatement prepareStatement(String sql)</code>	Creates a <code>PreparedStatement</code> object for sending parameterized SQL statements to the database.
<code>PreparedStatement prepareStatement(String sql, int resultSetType, int resultSetConcurrency)</code>	Creates a <code>PreparedStatement</code> object that will generate <code>ResultSet</code> objects with the given type and concurrency.

<code>void rollback()</code>	Undoes all changes made in the current transaction and releases any database locks currently held by this Connection object.
<code>void setAutoCommit(Boolean autoCommit)</code>	Sets this connection's auto-commit mode to the given state.

Executing Queries

To execute an SQL query, you have to use one of the following classes :

- Statement
- PreparedStatement
- CallableStatement

A Statement represents a general SQL statement without parameters. The method **createStatement()** creates a Statement object.

A PreparedStatement represents a precompiled SQL statement, with or without parameters. The method **prepareStatement(String sql)** creates a PreparedStatement object.

CallableStatement objects are used to execute SQL stored procedures. The method **prepareCall(String sql)** creates a CallableStatement object.

Executing a SQL statement with the Statement object, and returning a jdbc resultSet.

To execute a query, call an execute method from Statement such as the following :

- execute : Use this method if the query could return one or more ResultSet objects.
- executeQuery : Returns one ResultSet object.
- executeUpdate : Returns an integer representing the number of rows affected by the SQL statement. Use this method if you are using INSERT, DELETE, or UPDATE SQL statements.

Examples

```
ResultSet rs = stmt.executeQuery("SELECT * FROM book");
```

```
int result = stmt.executeUpdate("Update authors SET name = 'xxx' WHERE id = 1");
```

```
boolean bol = stmt.execute("DROP TABLE IF EXISTS DBTest");
```

ResultSet provides access to a table of data generated by executing a Statement.

The table rows are retrieved in sequence.

A ResultSet maintains a cursor pointing to its current row of data.

The next() method is used to successively step through the rows of the tabular results.

Examples :

```
Statement stmt = conn.prepareStatement();
ResultSet rs = stmt.executeQuery("Select * from student");
while(rs.next())
```

```

{
    //access resultset data
}

```

To access these values, there are getXXX() methods where XXX is a type for example, getString(), getInt() etc.

There are two forms of the getXXX methods:

- i. Using columnName: getXXX(String columnName)
- ii. Using columnNumber: getXXX(int columnNumber)

Example

```

rs.getString("studname")); rs.getString(1);
//where name appears as column 1 in the ResultSet

```

Using PreparedStatement

These are precompiled sql statements. For parameters, the SQL commands in a PreparedStatement can contain **placeholders** which are represented by '?' in the SQL command.

Example

```

String sql = "UPDATE authors SET name = ? WHERE id = ?";PreparedStatement ps =
conn.prepareStatement(sql);

```

Before the sql statement is executed, the placeholders have to be replaced by actual values. This is done by calling a **setXXX(int n, XXX x)** method,

where XXX is the appropriate type for the parameter

For example, **setString**, **setInt**, **setFloat**, **setDate** etc,

n is the placeholder number and x is the value which replaces the placeholder.

```

String sql = "UPDATE authors SET name = ? WHERE id = ?";
PreparedStatement ps = conn.prepareStatement(sql);
ps.setString(1,'abc'); //assign 'abc' to first placeholder
ps.setInt(2,123); //assign '123' to second placeholder

```

ResultSet Scroll Types and Concurrency

The scroll type indicates how the cursor moves in the ResultSet. The concurrency type affects concurrent access to the resultset. The types are given in the table below.

Scroll Type	
TYPE_FORWARD_ONLY	The result set is not scrollable.
TYPE_SCROLL_INSENSITIVE	The result set is scrollable but not sensitive to database changes.
TYPE_SCROLL_SENSITIVE	The result set is scrollable and sensitive to database changes.
Concurrency Type	
CONCUR_READ_ONLY	The result set cannot be used to update the database.
CONCUR_UPDATABLE	The result set can be used to update the database.

Example

```
Statement stmt = conn.createStatement  
(ResultSet.TYPE_SCROLL_SENSITIVE,  
ResultSet.CONCUR_UPDATABLE);
```

ResultSet Interface The ResultSet interface provides methods for retrieving and manipulating the results of executed queries.

Methods	Description
beforeFirst()	Default position. Puts cursor before 1st row of ResultSet.
first()	Puts cursor on 1st row of ResultSet.
last()	Puts cursor on last row of ResultSet.
afterLast()	Puts cursor after/beyond last row of ResultSet.
absolute (int pos)	Puts cursor at row number position where absolute (1) is a 1st row and absolute (-1) is last row of ResultSet.
relative (int pos)	Puts cursor at row no. position relative from current position.
next()	To move to the next row in ResultSet
previous()	To move to the previous row in ResultSet.
void close()	To close the ResultSet.
deleteRow()	Deletes the current row from the ResultSet and underlying database.
getRow()	Retrieves the current row number.
insertRow()	Inserts the contents of the insert row into the ResultSet object and into the database.
refreshRow()	Refreshes the current row with its most recent value in the database.
updateRow()	Updates the underlying database with the new contents of the current row of this ResultSet object.
getXXX(String columnName)	Retrieves the value of the designated column in the current row as a corresponding type in the Java programming language. XXX represents a type: Int, String, Float, Short, Long, Time etc.
moveToInsertRow()	Moves the cursor to the insert row.
close()	Disposes the ResultSet.
isFirst()	Tests whether the cursor is at the first position.
isLast()	Tests whether the cursor is at the last position.
isBeforeFirst()	Tests whether the cursor is before the first position.
isAfterLast()	Tests whether the cursor is after the last position.
updateXXX(int columnNumber, XXX value)	Updates the value of the designated column in the current row as a corresponding type in the Java programming language. XXX represents a type: Int, String, Float, Short, Long, Time etc.

DatabaseMetaData

This interface provides methods that tell you about the database for a given connection object.

Methods	Description
getDatabaseProductName()	Retrieves the name of this database product.
getDatabaseProductVersion()	Retrieves the version number of this database product.

getDriverName()	Retrieves the name of this JDBC driver.
getDriverVersion()	Retrieves the version number of this JDBC driver as a String.
getUserName()	Retrieves the user name as known to this database.
getCatalogs()	Retrieves the catalog names available in this database.
getSchemas(String catalog, String schemaPattern)	Retrieves the schema names available in this database.
getTables(String catalog, String schemaPattern, String tableNamePattern, String[] types)	Retrieves a description of the tables available in the given catalog.
getPrimaryKeys(String catalog, String schema, String table)	Retrieves a description of the given table's primary key columns.
getExportedKeys(String catalog, String schema, String table)	Retrieves a description of the foreign key columns that reference the given table's primary key columns (the foreign keys exported by a table).
getImportedKeys(String catalog, String schema, String table)	Retrieves a description of the primary key columns that are referenced by a table's foreign key columns (the primary keys imported by a table).
getColumns(String catalog, String schemaPattern, String tableNamePattern, String columnNamePattern)	Retrieves a description of table columns available in the specified catalog.
getProcedures(String catalog, String schemaPattern, String procedureNamePattern)	Retrieves a description of the stored procedures available in the given catalog.
getFunctions(String catalog, String schemaPattern, String functionNamePattern)	Retrieves a description of the system and user functions available in the given catalog.

Example

```
DatabaseMetaData dbmd = conn.getMetaData();
ResultSet rs = dbmd.getTables(null, null, null, new String[] { "TABLE" });
while (rs.next())
    System.out.println( rs.getString("TABLE_NAME"));
```

ResultSetMetaData

The ResultSetMetaData interface provides information about the structure of a particular ResultSet.

Methods	Description
getColumnCount()	Returns the number of columns in the current ResultSet object.
getColumnDisplaySize(int column)	Gives the maximum width of the column specified by the index parameter.
getColumnLabel(int column)	Gives the suggested title for the column for use in display and printouts.
getColumnName(int column)	Gives the column name associated with the column index.
getColumnTypeName(int column)	Gives the designated column's SQL type. isReadOnly(int column) Indicates whether the designated column is read-only.
isWritable(int column)	Indicates whether you can write to the designated column.
isNullable(int column)	Indicates the nullability of values in the designated column.

Self - Activity

Sample Program1 : Sample program to display employee data (empid, empname,salary)

```
import java.sql.*;
import java.io.*;
class JDBCdemo
{
    public static void main(String[] args) throws SQLException
    {
        Connection con = null;
        Statement stmt = null;
        ResultSet rs = null;
        try
        {Class.forName("org.postgresql.Driver");
            con = DriverManager.getConnection("jdbc:postgresql:empDB","postgres","");
            if(con==null)
                System.out.println("Connection failed ");
            else
            {
                System.out.println("Connection successful..");
                stmt = con.createStatement();
                rs = stmt.executeQuery("Select * from emp");
                while(rs.next())
                {System.out.print("EmpID = " + rs.getInt(1));
                    System.out.println("EmpName = " + rs.getString(2));
                    System.out.println("Salary = " + rs.getInt(3));
                }
                con.close();
            }
        }
        catch(Exception e)
        {System.out.println("ERROR"+e);
        }
    }
}
//end of main
// end of class
```

Sample Program2 : To perform insert and delete operations on employee table using PreparedStatement (Empid, Empname, Empsalary)

```
import java.sql.*;
import java.io.*;
class JDBCdemoOp
{
    public static void main(String[] args) throws SQLException
    {
        Connection con = null; Statement st = null;
        ResultSet rs = null;
        PreparedStatement ps1 = null, ps2=null; int eid, esal;
        String ename;
        BufferedReader br = new BufferedReader(new InputStreamReader(System.in));
        Class.forName("org.postgresql.Driver");
        con = DriverManager.getConnection("jdbc:postgresql:EmpDB","postgres","");
        st = conn.createStatement();
        ps1 = con.prepareStatement("Insert into employee values(?,?,?)");
        ps2 = con.prepareStatement("Delete employee where ID = ?");

        if(con!=null)
            System.out.println("Connection successful..");
        System.out.println("Enter the employee ID, employee name and
                           employee salary to be inserted ");
        eid = Integer.parseInt(br.readLine());
        ename = br.readLine();
        esal = Integer.parseInt(br.readLine());
        ps1.setInt(1,eid);
        ps1.setString(2,ename);
        ps1.setInt(3,esal);
        ps1.executeUpdate();

        System.out.println("Enter the employee ID to be deleted ");
        eid = Integer.parseInt(br.readLine());
        ps2.setInt(1,eid);
        ps2.executeUpdate();
        conn.close();
    } //end of main
} // end of class
```

Lab Assignments

SET A

1. Write a JDBC program to display all the details of the Person table in proper format on the screen. Create a Person table with fields as PID, name, gender, birth_year in PostgreSQL. Insert values in Person table.
2. Write a program to display information about the database and list all the tables in the database.

(Use DatabaseMetaData).

3. Write a program to display information about the ResultSet like number of columns available in the ResultSet and SQL type of the column. Use Person table. (Use ResultSetMetaData).
4. Write a JDBC program to display all the countries located in West Region. Create a table Country in PostgreSQL with fields (Name, continent, Capital, Region). Insert values in the table.

SET B

1. Write a JDBC program to insert the records into the table Employee(ID, name, salary) using PreparedStatement interface. Accept details of Employees from user.
2. Write a JDBC program to perform search operation on Person table.
 - i. Search all the person born in the year 1986.
 - ii. Search all the females born between 2000- 2005.
3. Write a JDBC program to update number_of_students of “BCA Science” to 1000. Create a table Course (Code, name, department, number_of_students). Insert values in the table.
4. Write a menu driven program to perform the following operations on District (Name, area, population) table.
 - i. Insert
 - ii. Modify
 - iii. Delete
 - iv. Search
 - v. View All
 - vi. Exit

SET C

1. Create a table Student with fields roll number, name, percentage using Postgresql. Insert values in the table. Write a JDBC program to display all the details of the student table in a tabular format on the screen. (Using Swing)
2. Create tables : Course (courseid, coursename, courseinstructor) Create a GUI based system for performing the following operations on the table:
Course : Add Course, View All students of a specific course

Assignment Evaluation

0: Not Done		1: Incomplete		2: Late Complete	
3: Needs Improvement		4: Complete		5: Well Done	

Signature of the instructor: _____ **Date:** _____

Assignment 4: Servlets

Objectives

- To understand server-side programming.
- Defining and executing servlets.
- To demonstrate the use of JSP

What are servlets?

Servlets are small programs that execute on the server side. Servlets are pieces of Java source code that add functionality to a web server

Servlet provides full support for sessions, a way to keep track of a particular user over time as a website's pages are being viewed. They also can communicate directly with a web server using a standard interface.

Servlets can be created using the `javax.servlet` and `javax.servlet.http` packages, which are a standard part of the Java's enterprise edition, an expanded version of the Java class library that supports large-scale development projects.

Running servlets requires a server that supports the technologies. Several web servers, each of which has its own installation, security and administration procedures, support Servlets. The most popular one is the Tomcat- an open source server developed by the Apache Software Foundation in cooperation with Sun Microsystems version 5.5 of Tomcat supports Java Servlet.

Getting Tomcat

The software is available a a free download from Apache's website at the address <http://jakarta.apache.org/tomcat>. Several versions are available: Linux users should download the rpm of Tomcat.

The `javax.servlet` package The important interfaces and classes are described in the table below:

The `javax.servlet` package

The important interfaces and classes are described in the table below.

Interface	Description
Servlet	A java servlet must implement the Servlet interface. This interface defines methods to initialize a servlet, to service requests, and to remove a servlet from the server. These are known as life-cycle methods.
ServletConfig	The ServletConfig interface is used by the server to pass configuration information to a servlet. Its methods are used by the servlet to retrieve this information.
taglib	Allows programmers to use new tags from tag libraries that encapsulate more complex functionality and simplify the coding of a JSP.
ServletRequest	The ServletRequest interface encapsulates a client request for service. It defines a number of methods for obtaining information about the server, requester, and request.

ServletResponse	The ServletResponse interface is used by a servlet to respond to a request by sending information back to the client.
ServletContext	The ServletContext interface defines the environment in which an applet is executed. It provides methods that are used by applets to access environment information.
SingleThreadModel	The SingleThreadModel interface is used to identify servlets that must be thread-safe. If a servlet implements this interface, the Web server will not concurrently execute the service() method of more than one instance of the servlet.

Class	Description
GenericServlet	The GenericServlet class implements the Servlet interface. You can subclass this class to define your own servlets.
ServletInputStream	The ServletInputStream class is used to access request information supplied by a Web client. An object of this class is returned by the getInputStream() method of the ServletRequest interface.
ServletOutputStream	The ServletOutputStream class is used to send response information to a Web client. An object of this class is returned by the getOutputStream() method of the ServletResponse interface.

The javax.servlet.http package

Interface	Description
HttpServletRequest	The HttpServletRequest interface extends the ServletRequest interface and adds methods for accessing the details of an HTTP request.
HttpServletResponse	The HttpServletResponse interface extends the ServletResponse interface and adds constants and methods for returning HTTP-specific responses
HttpSession	This interface is implemented by servlets to enable them to support browserserver sessions that span multiple HTTP request-response pairs. Since HTTP is a stateless protocol, session state is maintained externally using client-side cookies or URL rewriting. This interface provides methods for reading and writing state values and managing sessions.
HttpSessionContext	This interface is used to represent a collection of HttpSession objects that are associated with session IDs.

Class	Description
HttpServlet	Used to create HTTP servlets. The HttpServlet class extends the GenericServlet class.
Cookie	This class represents an HTTP cookie. Cookies are used to maintain session state over multiple HTTP requests. They are named data values that are created on the Web server and stored on individual browser clients. The Cookie class provides the method for getting and setting cookie values and attributes.

Servlet Life Cycle

A servlet's life cycle methods function similarly to the life cycle methods of applets.

- The init(ServletConfig) method is called automatically when a web server first begins

a servlet to handle the user's request. The `init()` method is called only once.

`ServletConfig` is an interface in the `javax.servlet` package, containing the methods to find out more about the environment in which a servlet is running.

- The servlet action is in the `service()` method. The `service()` method checks the HTTP request type (GET, POST, PUT, DELETE etc.) and calls `doGet()`, `doPost()`, `doPut()`, `doDelete()` etc. methods. A GET request results from normal request for a URL or from an HTML form that has no METHOD specified. The POST request results from an HTML form that specifically lists POST as the METHOD.
- The `destroy()` method is called when a web server takes a servlet offline.

Using Servlets

One of the main tasks of a servlet is to collect information from a web user and present something back in response. Collection of information is achieved using form, which is a group of text boxes, radio buttons, text areas, and other input fields on the web page. Each field on a form stores information that can be transmitted to a web server and then sent to a Java servlet. web browsers communicate with servers by using Hypertext Transfer Protocol(HTTP).

- Form data can be sent to a server using two kinds of HTTP requests: get and post. When web page calls a server using get or post, the name of the program that handles the request must be specified as a web address, also called uniform resource locator (URL). A get request affixes all data on a form to the end of a URL. A post request includes form data as a header and sent separately from the URL. This is generally preferred, and it's required when confidential information is being collected on the form.
- Java servlets handle both of these requests through methods inherited from the `HttpServlet` class: `doGet(HttpServletRequest, HttpServletResponse)` and `doPost(HttpServletRequest, HttpServletResponse)`. These methods throw two kinds of exceptions: `ServletException`, part of `javax.servlet` package, and `IOException`, an exception in the `java.io` package.
- The `getParameter(String)` method is used to retrieve the fields in a servlet with the name of the field as an argument. Using an HTML document a servlet communicates with the user.
- While preparing the response you have to define the kind of content the servlet is sending to a browser. The `setContentType(String)` method is used to decide the type of response servlet is communicating. Most common form of response is written using an HTML as: `setContentType("text/html")`.
- To send data to the browser, you create a servlet output stream associated with the browser and then call the `println(String)` method on that stream. The `getWriter()` method of `HttpServletResponse` object returns a stream. which can be used to send a response back to the client.

Example

```

import java.io.*;
import javax.servlet.* ;
import javax.servlet.http.*;
public class MyHttpServlet extends HttpServlet
{
    public void doGet(HttpServletRequest req,HttpServletResponse res)
        throws ServletException, IOException
    {
        // Use "req" to read incoming request
        // Use "res" to specify the HTTP response status
        //Use req.getParameter(String) or getParameterValues(String) to obtain
        parameters
        PrintWriter out = res.getWriter();//stream for output
        // Use "out" to send content to browser
    }
}

```

Request and Response methods

ServletRequest methods	Description
String getParameter(String name)	Obtains the value of a parameter sent to the servlet as part of a get or post request. The name argument represents the parameter name.
Enumeration getParameterNames()	Returns the names of all the parameters sent to the servlet as part of a post request.
String[]getParameterValues(String name)	For a parameter with multiple values, this method Returns an array of strings containing the values for a specified servlet parameter.
String getProtocol()	Returns the name and version of the protocol the request uses in the form protocol/majorVersion.minorVersion for example HTTP/1.
String getRemoteAddr()	Returns the Internet Protocol (IP) address of the client that sent the request.
String getRemoteHost()	Returns the fully qualified name of the client that sent the request.
String getServerName()	Returns the host name of the server that received the request.
int getServerPort()	Returns the port number on which this request was received.

HttpServletRequest methods	Description
Cookie[] getCookies()	Cookie[] getCookies() Returns an array of Cookie objects stored on the client by the server.
HttpSession getSession(boolean create)	Returns an HttpSession object associated with the client's current browsing session. This method can create an HttpSession object (True argument) if one does not already exist for the client.
String getServletPath()	Returns the part of this request's URL that calls the servlet

String getMethod()	Returns the name of the HTTP method with which this request was made for example GET, POST, or PUT.
String getQueryString()	Returns the query string that is contained in the request URL after the path.
String getRemoteUser()	Returns the login of the user making this request, if the user has been authenticated, or null if the user has not been authenticated.

ServletResponse methods	Description
ServletOutputStream getOutputStream()	Obtains a byte-based output stream for sending binary data to the client.
PrintWriter getWriter()	Obtains a character based output stream for sending text data (usually HTML formatted text) to the client.
void setContentType(String type)	Specifies the content type of the response to the browser. The content type is also known as MIME (Multipurpose Internet Mail Extension) type of the data. For examples, "text/html" , "image/gif" etc.
String setContentLength(int len)	Sets the length of the content body in the response In HTTP servlets, this method sets the HTTP Content-Length header

HttpServletResponse methods	Description
void addCookie(Cookie cookie)	Used to add a Cookie to the header of the response to the client.
void sendError(int ec	Sends an error response to the client using the specified status.
void sendError(int ec String msg)	Sends an error response to the client using the specified status code and descriptive message.
void sendRedirect(String url)	Sends a temporary redirect response to the client using the specified redirect location URL.
void setHeader(String name, String value)	Sets a response header with the given name and value.

Writing, Compiling and Running Servlet

- Type the first sample program of the self-activity section.
- After saving this servlet, compile it with the Java compiler as:

```
javac MyServlet.java.
```

- After compilation a class file with name MyServlet.class is created.

To make the servlet available, you have to publish this class file in a folder on your web server that has been designated for Java servlets. Tomcat provides the classes sub-folder to deploy this servlet's class file. Copy this class file in this classes sub-folder, which is

available on the path: tomcat/webapps/ WEB-INF/classes.

Now edit the web.xml file available under WEB-INF sub-folder with the following lines:

```
<servlet>
  <servlet-name>MyServlet</servlet-name>
  <servlet-class> MyServlet</servlet-class>
</servlet>

<servlet-mapping>
  <servlet-name> MyServlet </servlet-name>
  <url-pattern>/ MyServlet </url-pattern>
</servlet-mapping>
```

Repeat the above sequence of line to run every newly created servlet. Remember, these line lines must be placed somewhere after the <web-app> tag and before the closing </web-app> tag.

After adding these lines, save web.xml file. Restart the Tomcat service and run the servlet by loading its address with a web browser as: <http://localhost:8080/MyServlet>.

Using PostgreSQL – Database Connectivity tool with servlets

Java's Servlet also provides support for data handling using **PostgreSQL** database. For this you have to do few simple steps.

1. Copy the jar file mentioned in Database Connectivity assignment into the subfolder: tomcat/lib/common.
2. Edit the file .bash_profile of your login using command: vi .bash_profile.
3. Add the following line without removing any line.
4. export CLASSPATH=\$CLASSPATH:/\$HOME/tomcat/common/lib/<jar file> used in database connectivity assignment.
5. Example: if I have postgresql-9.3-1104.jdbc4.jar file, I will type the line as export CLASSPATH=\$CLASSPATH:/\$HOME/tomcat/common/lib/postgresql-9.3-1104.jdbc4.jar
6. Save this file. Logout from the terminal and re-login.
7. Create the table student(sno, sname) in your database. Insert few records into this table.

Session Handling

1. Using cookies
2. Using HttpSession class

1. Using Cookies

To keep the track of information about you and the features you want the site to display. This customization is possible because of a web browser features called cookies, small files containing information that a website wants to remember about a user, like username, number of visits, and other. The files are stored on the user's computer, and a website can read only the cookies on the user's system that the site has created. The default behavior of all the web browsers is to accept all cookies.

The `javax.servlet.http.Cookie` class allows us to create a cookie and send it to the browser. **The methods are:**

Method	Description
<code>int getMaxAge()</code>	Returns the maximum age of the cookie, specified in seconds. By default, -1 indicating the cookie will persist until browser shutdown.
<code>String getName()</code>	Returns the name of the cookie.
<code>String getValue()</code>	Returns the value of the cookie.
<code>void setMaxAge(int s)</code>	Sets the maximum age of the cookie in seconds.
<code>void setValue (String value)</code>	Assigns a new value to a cookie after the cookie is created.

- The `Cookie` class in the `javax.servlet.http` package supports cookies. To create a cookie, call the `Cookie(String,String)` constructor. The first argument is the name you want to give the `Cookie`, and the second is the cookie's value.
- To send a cookie, call the `addCookie(Cookie)` method of an `HttpServletResponse` object. You can add more than one cookie to a response.
- In a servlet, call the `getCookies()` method of an `HttpServletRequest` object to receive an array of `Cookie` objects. Use `getName()` and `getValue()` methods to find out about cookie.

2.HttpSession class

Servlet can retain the state of user through `HttpSession`, a class that represents sessions. There can be one session object for each user running your servlet.

- A user's session can be created or retrieved by calling the `getSession(Boolean)` method of the servlet's request object. Use an argument `true` if a session should be created when one doesn't already exist for the user.

Example

```
HttpSession hs=req.getSession(true);
```

```
public void doGet (HttpServletRequest req, HttpServletResponse res)
    throws ServletException, IOException
{
    HttpSession hs = req.getSession(true);

    // ....
}
```

- Objects held by session are called its attributes. Call the session's `setAttribute(String, Object)` method with two arguments: a name to give the attribute and the object.
- To retrieve an attribute, call the `getAttribute(String)` method with its name as the only argument. It returns the object, which must be cast from object to the desired class, or null if no attribute of that name exists.
- To remove an attribute when it's no longer needed, call `removeAttribute(String)` with its name as the argument.

Method	Description
Object getAttribute(String name)	Returns the object bound with the specified name in this session, or null if no object is bound under the name.
Enumeration getAttributeNames()	Returns an Enumeration of String objects containing the names of all the objects bound to this session.
long getCreationTime()	Returns the time when this session was created, measured in milliseconds since midnight January 1, 1970 GMT.
long getLastAccessedTime()	Returns the last time the client sent a request associated with this session, as the number of milliseconds since midnight January 1, 1970 GMT, and marked by the time the container received the request.
int getMaxInactiveInterval()	Returns the maximum time interval, in seconds, that the servlet container will keep this session open between client accesses
void RemoveAttribute(String name)	Removes the object bound with the specified name from this session.
void setAttribute(String name, Object value)	Binds an object to this session, using the name specified.
void setMaxInactiveInterval(int seconds)	Specifies the time, in seconds, between client requests before the servlet container will invalidate this session.
void invalidate()	Invalidates this session then unbinds any objects bound to it.
Boolean isNew()	Returns true if it is a new session.
String getId()	Returns a string containing the unique identifier assigned to this session.

Self Activity

Sample Program1 : Program for simple servlet.

```

import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
public class MyServlet extends HttpServlet
{
    public void service(HttpServletRequest req, HttpServletResponse res)
        throws ServletException, IOException
    {
        res.setContentType("text/html");
        PrintWriter pw = rs.getWriter();
        pw.println("<html>");
        pw.println("<body>");
        pw.println("<B> Welcome to My Servlet World... ");
        pw.println("</body>");
        pw.println("</html>");
        pw.close();}
}

```

After saving this servlet, compile it with the Java compiler as: `javac MyServlet.java`. Run the servlet using <http://localhost:8080/MyServlet>

Sample Program2: To read two numbers and return their Subtraction.

// Save the following code as Sub.html

```
<html>
<head>
<title>Subtraction of Two Number </title>
</head>
<body>
<form method="post" action="http://server-ip or localhost:8080/SubServlet">
  Enter the Number1 <input type="text" name="No1">
  Enter the Number2 <input type="text" name="No2">
  <br> <input type="Submit" value ="SUB" >
  <br> <input type="Reset" value ="CLEAR" >

</form>
</body>
</html>
```

// Save the following code as SubServlet.java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class SubServlet extends HttpServlet
{
    public void doGet(HttpServletRequest req,HttpServletResponse res)
        throws ServletException, IOException
    {
        int num1 = Integer.parseInt(req.getParameter("No1"));
        int num2 = Integer.parseInt(req.getParameter("No2"));

        int sub = num1 - num2;
        res.setContentType("text/html");
        PrintWriter pw = res.getWriter();
        pw.println("<h1> Subtraction </h1> <h3>"+sub+"</h3>");
        pw.close();
    }
}
```

Sample Program3 :For database handling using servlet

```

//Create a student table (sno, sname)
//The servlet displays all records from the student table on the client machine.
import java.io.*;
import java.text.*;
import java.util.*;
import javax.servlet.*;
import javax.servlet.http.*;
import java.sql.*;
public class ServletJdbc extends HttpServlet
{
    public void doGet(HttpServletRequest req, HttpServletResponse res) throws IOException,
ServletException
    {
        res.setContentType("text/html");
        PrintWriter out = res.getWriter();
        try
        {
            out.println("<html>");
            out.println("<body>");
            Class.forName("org.postgresql.Driver");
            out.println("<h1>Driver loaded</h1>");
            Connection c =
            DriverManager.getConnection("jdbc:postgresql:m2","postgres","");
            out.println("<h1>Connection created</h1>");
            Statement st=c.createStatement();
            ResultSet rs=st.executeQuery("select * from student");

            while(rs.next())
            {
                out.print("<h3>" + rs.getInt(1) + " – "
                + rs.getString(2) + "</h3>");
                out.println("<br>");
            }
        }
        catch(SQLException e)
        {
            out.println("ERROR" + e);
        }
        out.println("<h1>Hi! Manisha</h1>");
        out.println("</body>");
        out.println("</html>");
    }
}

```

Run this program as <http://server-ip:8080/ServletJdbc>

Sample Program 4 : For Add the cookies

//Save this program as AddCookie.java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class AddCookie extends HttpServlet
{
    public void doGet(HttpServletRequest req,HttpServletResponse res)
        throws ServletException, IOException
    {

        Cookie c1=new Cookie("Cookie1","1");
        res.addCookie(c1);
        res.setContentType("text/html");
        PrintWriter out = res.getWriter();
        out.print("Cookie added with value 1);
        Cookie c2=new Cookie("Cookie2","2");
        res.addCookie(c2);
        out.print("Cookie added with value 2);
        out.close();
    }
}
```

//Save this program as GetCookie.java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
public class GetCookie extends HttpServlet
{
    public void doGet(HttpServletRequest req,HttpServletResponse res)
        throws ServletException, IOException
    {
        Cookie [] c=req.getCookies();
        res.setContentType("text/html");
        PrintWriter out = res.getWriter();
        for(int i=0;i<c.length;i++)
            out.println("Cookie Name"+c[i].getName()); pw.close();
    }
}
```

Run this program as <http://server-ip:8080/AddCookie>

Run this program as <http://server-ip:8080/GetCookie>

Sample Program 5 : Program for Session using Servlet

//Save this program as DemoSession.java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
public class DemoSession extends HttpServlet
{
    String result1="success";
    String result2="failure";
    public void doGet(HttpServletRequest req,HttpServletResponse res)
        throws ServletException, IOException
    {
        HttpSession hs=req.getSession(true);
        String uname = req.getParameter("txt1");
        String pwd = req.getParameter("txt2");
        res.setContentType("text/html");
        PrintWriter pw=res.getWriter();
        if((uname.equals("covid"))&&(pwd.equals("covid19")))
        {
            pw.print("<a href=http://localhost:8080/NewInfo.html>
                Login Success</a>");
            hs.setAttribute("loginID",result1);
        }
        else
        {
            pw.print("<a href=http://localhost:8080/NewInfo.html> Kick Out</a>");
            hs.setAttribute("loginID",result2);
        }
        pw.close();
    }
}
```

```

<!--HTML File for NewInfo.html --> <html>
<head> <title></title>
</head>
<body> <form method="post" action="http://localhost:8080/SessionInfo">

<input type="Submit" value="Read Session Value">
</form>
</body>
</html>

```

//Save this program as SessionInfo.java

```

import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class SessionInfo extends HttpServlet
{
    String readloginid;
    public void doGet(HttpServletRequest req,HttpServletResponse res)
        throws ServletException, IOException
    {

        HttpSession hs = req.getSession(true);
        readloginid = hs.getId();
        res.setContentType("text/html");
        PrintWriter pw = res.getWriter();

        if(hs.getAttribute("loginID").equals("success"))
            pw.print("Your Session ID " + readloginid);
        else
            pw.print("<h1>Session Expired </h1>");
        pw.close();
    }
}

```

Create an html file for login and password and use `http://server-ip:8080/SessionDemo` in the Form Action tag.

SET A

1. Write a servlet program to display IP address, browser type, current date and time of server.
2. Write a servlet program to accept username and display message as “Have a Nice day!!!” with username.
3. Design a servlet to display “Welcome IP address of client” to first time visitor. Display “Welcome-back IP address of client” if the user is revisiting the page. (Use Cookies) (Hint: Use `req.getRemoteAddr()` to get IP address of client)

SET B

1. Design the table User (username, password) using Postgre Database. Design HTML login screen. Accept the user name and password from the user. Write a servlet program to accept the login name and password and validates it from the database you have created. If it is correct then display Welcome.html otherwise display Error.html.
2. Write a Servlet program to display number of times user has visited the page. (Use Session)

SET C

1. Write a servlet program to create a shopping mall. User must be allowed to do purchase from two pages. Each page should have a page total. The third page should display a bill, which consists of a page total of whatever the purchase has been done and print the total. (Use HttpSession)

Assignment Evaluation

0: Not Done		1: Incomplete		2: Late Complete	
3: Needs Improvement		4: Complete		5: Well Done	

Signature of the instructor: _____ **Date:** _____

Assignment 5: JSP

Objectives

- To demonstrate the use of JSP
- Defining and executing JSP

What is JSP?

JSP is Java Server Page, which is a dynamic web page and used to build dynamic websites. To run jsp, we need web server which can be tomcat provided by apache, it can also be jRun, jBoss(Redhat), weblogic (BEA) , or websphere(IBM).

JSP is dynamic file whereas Html file is static. HTML cannot get data from database or dynamic data. JSP can be interactive and communicate with database and controllable by programmer. It is saved by extension of .jsp. Each Java server page is compiled into a servlet before it can be used. This is normally done when the first request to the JSP page is made.

A JSP contains 3 important types of elements

1. Directives:- these are messages to the JSP container that is the server program that executes JSPs.
2. Scripting elements:- These enables programmers to insert java code which will be a part of the resultant servlet.
3. Actions:- Actions encapsulates functionally in predefined tags that programmers can embedded in a JSP.

JSP Directives

Directives are message to the JSP container that enable the programmer to specify page setting to include content from other resources & to specify custom tag libraries for use in a JSP.

Syntax

<%@ name attribute1="..." , attribute2="..."...%>

Directive	Description
page	Defines page settings for the JSP container to process.
include	Causes the JSP container to perform a translation-time insertion of another resource's content. The file included can be either static (HTML file) or dynamic (i.e., another tag file)
taglib	Allows programmers to use new tags from tag libraries that encapsulate more complex functionality and simplify the coding of a JSP.

Page Directive

The page directives specify global settings for the JSP in the JSP container. There can be many page directives, provided that there is only one occurrence of each attribute.

Syntax

```

<%@ page
[ language="java" ]
[ extends="package.class" ]
[ import="{package.class | package.*}, ..." ]
[ session="true|false" ]
[ buffer="none|8kb|sizekb" ]
[ autoFlush="true|false" ]
[ isThreadSafe="true|false" ]
[ info="text" ]
[ errorPage="relativeURL" ]
[ contentType="mimeType [ ; charset=characterSet ]" "text/html ; charset=ISO-8859-1" ]
[ isErrorPage="true|false" ]
[ pageEncoding="characterSet | ISO-8859-1" ] %>

```

Scripting Elements

1. Declarations

A declaration declares one or more variables or methods that you can use in Java code later in the JSP file.

Syntax

```
<%! Java declaration statements %>
```

Example

```
<%! private int count = 0; %> <%! int i = 0; %>
```

2. Expressions

An expression element contains a java expression that is evaluated, converted to a String, and inserted where the expression appears in the JSP file.

Syntax

```
<%= expression %>
```

Example

```
Name is <%= request.getParameter("name") %>
```

3. Scriptlet

A scriptlet contains a set of java statements which is executed. A scriptlet can have java variable and method declarations, expressions, use implicit objects and contain any other statement valid in java.

Syntax

```
<% statements %>
```

Example

```
<% String name = request.getParameter("userName");
out.println("Hello " + name); %>
```

Implicit objects used in JSP

Implicit object	Description
application	A javax.servlet.ServletContext object that represents the container in which the JSP executes. It allows sharing information between the jsp page's servlet and any web components within the same application.
config	A javax.servlet.ServletConfig object that represents the JSP configuration options. As with servlets, configuration options can be specified in a Web application descriptor (web.xml). The method getInitParameter() is used to access the initialization parameters.
exception	A java.lang.Throwable object that represents an exception that is passed to a JSP error page. This object is available only in a JSP error page.
out	A javax.servlet.jsp.JspWriter object that writes text as part of the response to a request. This object is used implicitly with JSP expressions and actions that insert string content in a response.
page	An Object that represents the current JSP instance.
pageContext	A javax.servlet.jsp.PageContext object that provides JSP programmers with access to the implicit objects discussed in this table.
request	An object that represents the client request and is normally an instance of a class that implements HttpServletRequest. If a protocol other than HTTP is used, this object is an instance of a subclass of javax.servlet.ServletRequest. It uses the getParameter() method to access the request parameter.
response	An object that represents the response to the client and is normally an instance of a class that implements HttpServletResponse (package javax.servlet.http). If a protocol other than HTTP is used, this object is an instance of a class that implements javax.servlet.ServletResponse.
session	A javax.servlet.http.HttpSession object that represents the client session information. This object is available only in pages that participate in a session.

To run JSP files: all JSP code should be copied (Deployed) into webapps folder in the tomcat server. To execute the file, type: `http://server-ip:8080/Programname.jsp`

Self Activity

Sample Program1 : Simple display on browser.

```
/* type this as first.jsp */
<html>
<body>
  <% out.print("DREAMS Don't work UNLESS YOU DO!"); %>
</body>
</html>
```

Sample Program2 : To display current date.

```
<%@ page language="java" import="java.util.*" %>
<html>
  <body> Current Date time: <%=new java.util.Date()%>
</body>
</html>
```

Sample Program3 : To multiplication of two numbers “MultNumbers.jsp”

```
<%@ page language="java"%>
<html>
<head>
<title>Add number program in JSP </title>
</head>
<body>
<form method = “post” action = “MultNumbers.jsp”>
  Enter Number 1 <input type =”text” name = “No1”>
  Enter Number 2 <input type =”text” name = “No2”>
  <input type=”submit” value=”RESULT”/>
  <%
    int p = Integer.parseInt(request.getParameter(“No1”));
    int q = Integer.parseInt(request.getParameter(“No2”));
    int result = p * q;
    out.print("Multiplication of p and q :"+result);
  %>
```

Lab Assignments

SET A

1. Create a JSP page to accept a number from user and display it in words: Example: 123 – One Two Three.
2. Write a JSP program to check whether given number is Perfect or not.
3. Write a JSP program to perform Arithmetic operations such as Addition, Subtraction, Multiplication and Division. Design a HTML to accept two numbers in text box and radio buttons to display operations. On submit display result as per the selected operation on next page using JSP.

SET B

1. Create a JSP page, which accepts user name in a text box and greets the user according to the time on server side.

Example: If user name is Admin

Output: If it is morning then display message in red color as,
Good morning Admin

Today’s date: dd/mm/yyyy format

Current time: hh:mm:ss format

If it is afternoon then display message in green color as,
Good afternoon Admin
Today's date: dd/mm/yyyy format
Current time: hh:mm:ss format

If it is evening then display message in blue color as,
Good evening Admin
Today's date: dd/mm/yyyy format
Current time: hh:mm:ss format (Hint: To display date and time use
GregorianCalendar and Calendar class)

2. Write a JSP program to display number of times user has visited the page. (Use cookies)

SET C

1. Write a JSP program to display details of products from database in tabular format.
Design Product table in database as (pid, pname, qty, price)

Assignment Evaluation

0: Not Done		1: Incomplete		2: Late Complete	
3: Needs Improvement		4: Complete		5: Well Done	

Signature of the instructor: _____ **Date:** _____